

MRP Project Proposal

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Stock Market Regime Detection and Forecasting using Hidden Markov Models and LSTM Networks

1. Introduction

A. Background

Financial markets are complex and often switch between different conditions such as bullish, bearish, or high-volatility periods. These shifts are usually referred to as market regimes. Traditional models like ARIMA can capture trends but struggle with sudden changes, while deep learning models like LSTM capture time dependencies but do not explicitly model regime changes.

This project explores a hybrid approach where Hidden Markov Models (HMM) are used to identify hidden market regimes, and LSTM networks are used to forecast prices. The idea is that combining both can better capture market structure and improve prediction performance.

B. Research Questions

- Can HMM effectively identify hidden market regimes in financial data?
- Does incorporating regime information improve forecasting accuracy?
- How does a hybrid HMM-LSTM model compare to ARIMA and standalone LSTM?

C. Variables

Dependent Variable: Future stock price movement or returns

Independent Variables:

- Historical prices
- Technical indicators such as returns and volatility
- Regime states from HMM

2. Literature Review

Recent research has explored combining statistical models with deep learning for financial forecasting. Traditional methods like ARIMA are limited in capturing non-linear patterns. More

recent work uses HMM for regime detection and LSTM for sequence modeling, showing that incorporating regime-switching behavior can improve predictions.

Some studies also introduce variations like attention-based LSTM or hybrid frameworks, but results vary depending on the dataset and implementation. This project builds on these ideas by comparing hybrid and standalone models in a consistent experimental setup.

3. Exploratory Data Analysis

Before modeling, basic analysis will be performed on the dataset, including:

- Distribution of returns
- Volatility patterns over time
- Correlation between features
- Time series visualization

This step helps understand the structure of the data and guides feature selection.

4. Methodology and Experiments

A. Aim of Study

The goal is to evaluate whether a hybrid HMM-LSTM model improves forecasting accuracy compared to traditional and standalone models.

B. Modeling Approach

Baseline Models: ARIMA, standalone LSTM

Proposed Model: HMM for regime detection combined with LSTM for forecasting

C. Experimental Design

Data will be split into training and testing sets, using a 70/30 split.

K-fold cross-validation will be used for robustness. Multiple runs will be conducted to reduce randomness in results.

D. Evaluation Metrics

Mean Squared Error (MSE)

Mean Absolute Error (MAE)

Directional accuracy

E. Hypothesis and Statistical Testing

H0: The hybrid model does not improve performance.

H1: The hybrid model improves performance.

A paired t-test will be used with a significance level of 0.05 to validate the results.

5. Expected Results

It is expected that the hybrid model will perform better, especially during volatile periods, as it can adapt to different market regimes. However, performance will depend on how well the regimes are identified and used in the model.

6. Conclusion and Future Work

This project aims to combine HMM and LSTM to improve financial forecasting. The results will provide insight into whether regime-aware modeling leads to better predictions.

Future work could include:

- Adding more financial indicators

- Testing on different assets

- Exploring more advanced models

7. Reproducibility

All code and experiments will be shared on GitHub to ensure results can be reproduced.

8. Dataset Sources

The empirical analysis will utilize publicly accessible historical records from Yahoo Finance. Daily historical data will be extracted via Yahoo Finance Premium/Gold, spanning approximately three decades where data availability permits. This extensive temporal range is essential for capturing a diverse array of market environments, including sustained bullish and bearish trends, systemic financial crises, varying inflationary cycles, and regimes characterized by extreme volatility.

1. S&P 500 / Equity Market Data

Link: <https://finance.yahoo.com/quote/%5EGSPC/history/>

Description: The S&P 500 Index serves as the primary benchmark for the equity market.

Representing the performance of 500 large-cap U.S. firms, it provides a comprehensive reflection of the broader stock market. Daily pricing data will be leveraged to derive technical features including returns, volatility, and trend-based indicators.

2. SPDR S&P 500 ETF Trust (SPY)

Link: <https://finance.yahoo.com/quote/SPY/history/>

Description: As an exchange-traded fund tracking the S&P 500, SPY provides a tradable proxy for equity markets by including both price and liquidity (volume) data. This allows for validation of whether model results for the theoretical index translate consistently to real-world investable assets.

3. CBOE Volatility Index (VIX)

Link: <https://finance.yahoo.com/quote/%5EVIX/history/>

Description: Often utilized as a "fear gauge," the VIX quantifies expected market uncertainty. It will be incorporated as an independent feature to assist the HMM in distinguishing between stable, high-volatility, and crisis-driven market regimes.

4. 10-Year Treasury Yield Index (TNX)

Link: <https://finance.yahoo.com/quote/%5ETNX/history/>

Description: The 10-Year Treasury Yield will be included as a macroeconomic variable. Since interest rate fluctuations influence cross-asset performance, this dataset helps model the broader financial conditions that often precipitate regime shifts.

5. Gold Data

Link: <https://finance.yahoo.com/quote/GC%3DF/history/>

Description: Gold futures represent the precious metals market within this study. As a traditional safe-haven asset, gold allows for testing the HMM-LSTM framework's ability to detect regimes in assets that exhibit decorrelation with equity markets during distress.

6. Oil Data

Link: <https://finance.yahoo.com/quote/CL%3DF/history/>

Description: Crude oil futures will represent the energy commodity sector. Given that oil prices are sensitive to geopolitical events and supply-demand shocks, this dataset is valuable for evaluating regime detection under commodity-driven pressures.

7. Bitcoin Data

Link: <https://finance.yahoo.com/quote/BTC-USD/history/>

Description: Bitcoin will be utilized to model the cryptocurrency asset class. Characterized by extreme volatility and 24/7 trading, it provides a rigorous test case for the robustness of the hybrid framework in highly unstable environments.

Data Fields

The following daily metrics will be extracted across all datasets where available:

Open, High, Low, and Close prices
Adjusted Close price
Trading Volume

These raw inputs will be processed to construct features such as log returns, rolling standard deviation, moving averages, and the regime labels generated by the HMM.

Data Usage

A unified modeling pipeline will be applied to all asset classes. Data will undergo cleaning and transformation into standardized return and volatility features. Subsequently, HMMs will be deployed to uncover hidden states within each market. These regime-aware features will then serve as inputs for LSTM networks to forecast price movements. Comparing performance across diverse markets like equities and commodities will validate the model's generalizability.

9. References

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